LECTURE SERIES · DATABASE SYSTEMS

SQL Basics

A Beginner's Guide to Structured Query Language

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Introduction · Create · Insert · Update · Delete · Select · Where · Like · Joins

INTRODUCTION

What is SQL?

SQL (Structured Query Language) is the standard language used to store, retrieve, and manage data held in **relational databases.**

Declarative

You describe what you want — the engine figures out how.

Universal

Powers MySQL, PostgreSQL, SQL Server, Oracle & SQLite.

Organized

Data lives in tables of rows (records) and columns (fields).

TODAY'S ROADMAP

- **DDL** *Define structure*
  `CREATE`
- **DML** *Change data*
  `INSERT · UPDATE · DELETE`
- **Query** *Read data*
  `SELECT · WHERE · LIKE`
- **Combine** *Relate tables*
  `JOINS`

DDL · DATA DEFINITION

CREATE — Define a Table

The CREATE TABLE statement builds a new table by naming its columns and the type of data each will hold.

Tip: PRIMARY KEY uniquely identifies every row.

Common types: `INT` · `VARCHAR(n)` · `FLOAT` · `DATE`

schema.sql

```sql
CREATE TABLE Students (
    id      INT PRIMARY KEY,
    name    VARCHAR(50),
    dept    VARCHAR(20),
    cgpa    FLOAT
);

-- a table named Students with 4 columns
```

DML · DATA MANIPULATION

INSERT — Add Records

INSERT INTO adds a new row of data. List the columns, then supply matching VALUES in the same order.

Note: Wrap text/date values in single quotes.

insert.sql

```
INSERT INTO Students (id, name, dept,
cgpa)
VALUES (1, 'Rakib', 'CSE', 3.90);
```

Result — one row added:

id	name	dept	cgpa
1	Rakib	CSE	3.90

DML · DATA MANIPULATION

UPDATE — Modify Records

UPDATE changes values in existing rows. SET assigns the new value; WHERE decides which rows are affected.

Careful! Omit WHERE and every row gets updated.

update.sql

```sql
UPDATE Students
SET cgpa = 3.95
WHERE id = 1;
```

Before → After:

id	name	cgpa
1	Rakib	3.90 → 3.95

DML · DATA MANIPULATION

DELETE — Remove Records

DELETE FROM removes rows from a table. The WHERE clause targets exactly which rows should be deleted.

Danger! No WHERE = the whole table is emptied.

delete.sql

```
DELETE FROM Students
WHERE id = 1;
```

Best practice: Run a SELECT with the same WHERE first to preview the rows you are about to remove.

QUERYING

SELECT — Retrieve Data

SELECT reads data from a table. Name the columns you want, or use * for all of them.

select.sql

```sql
-- specific columns
SELECT name, dept, cgpa
FROM Students;

-- every column
SELECT * FROM Students;
```

Query output:

id	name	dept	cgpa
1	Rakib	CSE	3.90
2	Alifa	CSE	3.85
3	Nadia	EEE	3.70

Tip: Add ORDER BY cgpa DESC to sort results.

QUERYING

WHERE — Filter Rows

WHERE keeps only the rows that meet a condition.
Combine conditions with AND, OR and NOT.

where.sql

```sql
SELECT * FROM Students
WHERE dept = 'CSE'
  AND cgpa > 3.5;
```

Comparison operators

`=` `>` `<` `>=` `<=` `<>` `AND` `OR`

`NOT` `BETWEEN` `IN`

Reads as: “Give me all CSE students whose CGPA is above 3.5.”
Conditions are checked row by row.

QUERYING

LIKE — Pattern Matching

LIKE searches text using wildcards instead of an exact match — perfect for “starts with”, “ends with” or “contains”.

like.sql

```
SELECT * FROM Students
WHERE name LIKE 'R%';
```

Meaning: Finds every name beginning with the letter R.

Wildcards

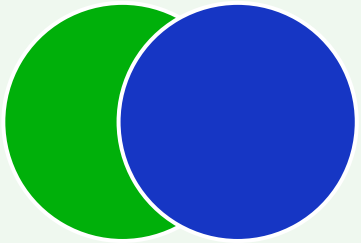
Pattern	Matches
'R%'	Rakib, Rana, Rio
'%b'	Rakib, Sohab
'_a%'	Nadia, Rana
'%CSE%'	contains CSE

% = any number of characters · **_** = exactly one character

RELATING DATA

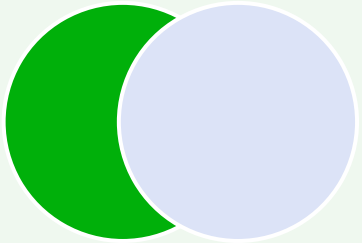
SQL JOINS — Combine Tables

A JOIN merges rows from two tables using a related column. The join type decides which unmatched rows are kept.



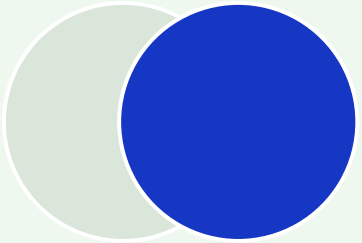
INNER

only matching rows



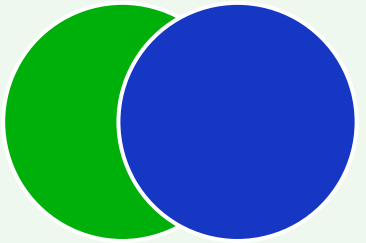
LEFT

all left + matches



RIGHT

all right + matches



FULL

all rows, both sides

join.sql

```sql
SELECT s.name, d.dept_name   FROM Students s
INNER JOIN Departments d   ON s.dept_id = d.id;
```

KEY TAKEAWAYS

Recap

CREATE

Define a table & its columns

INSERT

Add new rows of data

UPDATE / DELETE

Change or remove rows (use WHERE!)

SELECT

Read data from tables

WHERE / LIKE

Filter rows & match patterns

JOINS

Combine related tables

Thank You

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